

Schemes

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1 Pushforward

Lemma 1.1. The direct image of a (pre)sheaf on X is a (pre)sheaf on Y .

Proof. Let $\pi : X \rightarrow Y$ be a continuous map.

Let $\mathcal{F}$ be a sheaf on X .

Let's show that the pushforward sheaf on Y satisfies the sheaf axioms.

Let $V_1 \subseteq V_2$ in Y . Then, $\pi^{-1}(V_1) \subseteq \pi^{-1}(V_2)$ in X . Thus, we have an induced restriction map using the restriction map from X .

Similarly, we have that the restriction triangle is commutative using the commutativity on X .

Thus, $\pi_*\mathcal{F}$ is a presheaf.

Now, let V be an open subset of Y and let $\{V_i\}_{i \in I}$ be an open cover of V . Notice that $\{\pi^{-1}(V_i)\}_{i \in I}$ is an open cover of $\pi^{-1}(V)$.

Let $f_1, f_2 \in \pi_*\mathcal{F}(V) = \mathcal{F}(\pi^{-1}(V))$. But then this is simply the gluability axiom and identity axioms on X . $\square$

Lemma 1.2. Let $\pi : X \rightarrow Y$ be a continuous map and $\mathcal{F}$ be a sheaf of sets, rings or A -modules on X . Let $p \in X$ and $q \in Y$ such that $\pi(p) = q$. Then, the pushforward sheaf induces a map on the stalks

Proof. Let (g, V) be a germ of $(\pi_*\mathcal{F})_q$.

Let $U = \pi^{-1}(V)$.

Then, $q \in V$ and $g \in \pi_*\mathcal{F}(V) = \mathcal{F}(U)$.

Map (g, V) to (g, U) .

We need to show that this assignment is well-defined. $\square$

Let's now prove the same statement using the colimit definition of stalks.

Proof. $\square$